

Title Page

Predicting and Presenting Task Difficulty for Crowdsourcing Food Rescue Platforms

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Table of Contents

1. Introduction
2. Background 2.1 Food Rescue Platforms 2.2 Volunteer Engagement and Retention 2.3 Machine Learning Concepts
3. Contributions 3.1 Research Contributions 3.2 My Contributions
4. Implementation 4.1 Programming Languages and Tools 4.2 Data Preprocessing and Debugging 4.3 Model Development and Comparison 4.4 AWS Integration (Future Work)
5. Outcome
6. Issues and Reflections 6.1 Personal Learning and Development 6.2 Challenges Encountered
7. Bibliography

Introduction

This spring semester, I engaged in an AI research project supervised by Dr. Ryan Shi at the University of Pittsburgh, in collaboration with Carnegie Mellon University, UIUC, and Food Rescue Hero. Our research targets volunteer retention by predicting the difficulty of tasks in food rescue platforms. This paper elaborates on the project's background, my contributions, detailed implementations, outcomes, and personal reflections.

Background

2.1 Food Rescue Platforms

Food waste and food insecurity are two problems that co-exist worldwide. A major force to combat food waste and insecurity, food rescue platforms (FRP) match food donations to low-resource communities. Since they rely on external volunteers to deliver the food, communicating rescue task difficulty to volunteers is very important for volunteer engagement and retention. We develop a hybrid model with tabular and natural language data to predict the difficulty of a given rescue trip, which significantly outperforms baselines in identifying easy and hard rescues. Furthermore, using storyboards, we conducted interviews with different stakeholders to understand their perspectives on how to integrate such predictions into volunteers' workflow. Motivated by our findings, we developed three explanation methods to generate interpretable insights for volunteers to better understand the predictions. The results from this study are in the process of being adopted at Food Rescue Hero, a large FRP serving over 25 cities across the United States.

2.2 Volunteer Engagement and Retention

Volunteer retention is crucial for the success of FRPs. Despite their importance, FRPs face significant challenges in maintaining a stable volunteer workforce. Volunteers often experience high attrition rates due to negative first-time experiences, such as confusing pickup locations, unexpected difficulties during the task, or prolonged travel times. Such negative initial experiences can greatly discourage new volunteers from participating in future rescues, posing a major operational challenge for these platforms. Ensuring a positive initial experience is therefore pivotal.

To address these challenges, FRPs strive to strategically match tasks according to volunteer experience levels—simpler tasks are directed towards new volunteers, while complex tasks are recommended for seasoned volunteers. Effective matching can reduce early attrition, keeping volunteers engaged and motivated. Thus, clearly communicating task difficulty upfront can substantially improve volunteer retention rates.

In our research, we collaborated closely with Food Rescue Hero, a large-scale FRP operating across 25 cities in the United States with a volunteer base of approximately 45,000 people. Food Rescue Hero delivers millions of pounds of food annually, which highlights the critical role volunteers play in their operational success. Our collaboration specifically targeted the development of an AI system capable of accurately predicting task difficulty, thereby directly contributing to improved volunteer experiences and retention.

2.3 Machine Learning Concepts

The core concepts in our research involve supervised learning, natural language processing (NLP), and pseudo-labeling. Supervised learning is a type of machine learning where a model is

trained on labeled data. In our project, we trained models to predict task difficulty labels categorized by expert evaluations.

Natural language processing played a crucial role due to the presence of textual data—volunteer feedback. NLP techniques, particularly BERT (Bidirectional Encoder Representations from Transformers), were leveraged to process and interpret volunteer comments, which are critical indicators of task difficulty.

Pseudo-labeling emerged as a pivotal method in our project due to the scarcity of manually labeled data. Expert labeling of task difficulty is resource-intensive and expensive. Thus, we utilized a two-step BERT training method: first, the BERT model was pre-trained on volunteer-provided ratings, which were abundant but less directly indicative of task difficulty.

Subsequently, this pre-trained model was fine-tuned on a smaller set of expert-labeled data to predict task difficulty explicitly. This method allowed us to generate additional pseudo-labels, significantly expanding our training data.

Another key concept in our research was the evaluation metric ROC-AUC (Receiver Operating Characteristic - Area Under the Curve). ROC-AUC measures the ability of a model to distinguish between classes—in our case, easy and difficult tasks. ROC-AUC scores range from 0.5, equivalent to random guessing, to 1.0, which represents perfect prediction. This metric is particularly valuable in our research context, as it provides a balanced measure that accounts for the often-imbalanced nature of rescue task datasets.

Beyond ROC-AUC, we also utilized additional metrics such as F1 scores and precision-recall AUC (PR-AUC) to comprehensively evaluate model performance, ensuring the reliability and robustness of our predictions across diverse scenarios and volunteer experiences.

Contributions

3.1 Research Contributions

We developed a hybrid model using numeric and textual data. Given limited manually labeled data, we utilized a BERT model trained initially on user ratings, generating pseudo-labels. The final model performed strongly with a ROC-AUC of 0.710 for easy rescues and 0.685 for difficult rescues.

3.2 My Contributions

My role primarily involved debugging, refining data preprocessing, verifying data integrity, conducting comparative model analyses, and examining the temporal stability and efficiency of predictions.

Implementation

4.1 Programming Languages and Tools

First, we worked in Python and Jupyter, which gave us a flexible way to run and test code interactively, especially with all the data science libraries involved.

For data storage, we used PostgreSQL, and accessed it through SQLAlchemy and psycopg2—these let us load rescue records, volunteer comments, and annotations directly into Python.

On the machine learning side, we used scikit-learn for traditional tabular models like Random Forest and LightGBM. We also handled preprocessing steps like imputation and evaluated with metrics like ROC AUC and F1 score.

For the text data, we used Hugging Face Transformers—specifically models like BERT and DistilBERT—to analyze volunteer comments and generate difficulty labels. Once we had those pseudo-labels, we combined them with numeric features and trained models like LightGBM for the final predictions.

And finally, tools like matplotlib, seaborn, and tqdm helped us visualize model performance and track progress throughout the pipeline. All of this was tied together with collaboration on Slack, which made communication really smooth.

4.2 Data Preprocessing and Debugging

First, I debugged our existing code to ensure everything could compile and run smoothly. This included fixing library incompatibilities, refining data preprocessing steps, and confirming that both numeric and geographic features were properly ingested. By doing this, I made sure the foundation for model training was solid.

4.3 Model Development and Comparison

Next, I moved on to model development and comparison. I experimented with several tabular models and then I fine-tuned their parameters using cross-validation. I incorporated time-based

analysis to observe how each model’s predictions changed or trended over time. Ultimately, we wanted to identify which model had the most reliable performance not only in terms of accuracy or AUC scores, but also in terms of efficiency and stability. I systematically explored three main modeling approaches, each with rigorous performance evaluations:

4.3.1 Baseline 1

Split	Model	Accuracy	F1	ROC-AUC	PR-AUC
Easy	lgb	0.914 ± 0.000	0.066 ± 0.000	0.692 ± 0.000	0.142 ± 0.000
	rf	0.909 ± 0.003	0.114 ± 0.051	0.686 ± 0.034	0.159 ± 0.011
	lr	0.891 ± 0.000	0.100 ± 0.000	0.554 ± 0.000	0.107 ± 0.000
	svm	0.387 ± 0.355	0.105 ± 0.051	0.528 ± 0.069	0.081 ± 0.009
	mlp	0.732 ± 0.305	0.047 ± 0.072	0.567 ± 0.055	0.101 ± 0.017
	knn	0.922 ± 0.000	0.000 ± 0.000	0.481 ± 0.000	0.134 ± 0.000
Hard	lgb	0.909 ± 0.000	0.043 ± 0.000	0.405 ± 0.000	0.031 ± 0.000
	rf	0.877 ± 0.055	0.019 ± 0.024	0.417 ± 0.037	0.031 ± 0.003
	lr	0.566 ± 0.000	0.036 ± 0.000	0.393 ± 0.000	0.029 ± 0.000
	svm	0.223 ± 0.369	0.059 ± 0.029	0.641 ± 0.000	0.049 ± 0.000
	mlp	0.489 ± 0.360	0.046 ± 0.025	0.428 ± 0.053	0.031 ± 0.004
	knn	0.341 ± 0.000	0.044 ± 0.000	0.386 ± 0.000	0.029 ± 0.000

Baseline 1 involved purely tabular data without text features. The models tested included LightGBM, Random Forest, Logistic Regression, SVM, MLP, and KNN. The results showed moderate performance, notably with LightGBM achieving the highest ROC-AUC of 0.692 for easy tasks and 0.405 for hard tasks. However, other metrics such as accuracy and F1 scores were disappointingly low, highlighting the limitations of tabular-only data. The model which relies only on tabular data can not on their own to make accurate predictions. Once we include comment text and generate pseudo-labels from BERT, we see clear performance improvement.

4.3.2 Baseline 2

Baseline 2 (Raw Split) — <i>Easy</i>				
Model	Accuracy	F1	AUC	PR
LGB	0.638 ± 0.000	0.231 ± 0.000	0.703 ± 0.000	0.140 ± 0.000
RF	0.784 ± 0.000	0.252 ± 0.000	0.590 ± 0.000	0.110 ± 0.000
LR	0.828 ± 0.000	0.205 ± 0.000	0.571 ± 0.000	0.110 ± 0.000
SVM	0.466 ± 0.000	0.199 ± 0.000	0.614 ± 0.000	0.091 ± 0.000
MLP	0.573 ± 0.000	0.159 ± 0.000	0.485 ± 0.000	0.086 ± 0.000
KNN	0.903 ± 0.000	0.226 ± 0.000	0.518 ± 0.000	0.187 ± 0.000
Baseline 2 (Raw Split) — <i>Hard</i>				
Model	Accuracy	F1	AUC	PR
LGB	0.048 ± 0.000	0.074 ± 0.000	0.356 ± 0.000	0.028 ± 0.000
RF	0.043 ± 0.000	0.074 ± 0.000	0.433 ± 0.000	0.032 ± 0.000
LR	0.252 ± 0.000	0.079 ± 0.000	0.397 ± 0.000	0.029 ± 0.000
SVM	0.120 ± 0.000	0.078 ± 0.000	0.428 ± 0.000	0.031 ± 0.000
MLP	0.093 ± 0.000	0.074 ± 0.000	0.352 ± 0.000	0.028 ± 0.000
KNN	0.758 ± 0.000	0.062 ± 0.000	0.390 ± 0.000	0.030 ± 0.000

Baseline 2 incorporated text data using a fine-tuned BERT model to generate pseudo-labels.

Despite expecting improvement, the metrics indicated significant shortcomings. The ROC-AUC scores remained insufficient, particularly for hard tasks (only 0.356), far below expectations compared to values reported in previous papers.

4.3.3 Final Algorithm

Final Algorithm (Raw Split) – Easy				
Model	Accuracy (mean ± std)	F1 (mean ± std)	AUC (mean ± std)	PR (mean ± std)
LGB	0.638 ± 0.000	0.231 ± 0.000	0.703 ± 0.000	0.140 ± 0.000
RF	0.784 ± 0.000	0.252 ± 0.000	0.590 ± 0.000	0.110 ± 0.000
LR	0.828 ± 0.000	0.205 ± 0.000	0.571 ± 0.000	0.110 ± 0.000
SVM	0.466 ± 0.000	0.199 ± 0.000	0.614 ± 0.000	0.091 ± 0.000
MLP	0.573 ± 0.000	0.159 ± 0.000	0.485 ± 0.000	0.086 ± 0.000
KNN	0.903 ± 0.000	0.226 ± 0.000	0.518 ± 0.000	0.187 ± 0.000
Final Algorithm (Raw Split) – Hard				
Model	Accuracy (mean ± std)	F1 (mean ± std)	AUC (mean ± std)	PR (mean ± std)
LGB	0.048 ± 0.000	0.074 ± 0.000	0.356 ± 0.000	0.028 ± 0.000
RF	0.043 ± 0.000	0.074 ± 0.000	0.433 ± 0.000	0.032 ± 0.000
LR	0.252 ± 0.000	0.079 ± 0.000	0.397 ± 0.000	0.029 ± 0.000
SVM	0.120 ± 0.000	0.078 ± 0.000	0.428 ± 0.000	0.031 ± 0.000
MLP	0.093 ± 0.000	0.074 ± 0.000	0.352 ± 0.000	0.028 ± 0.000
KNN	0.758 ± 0.000	0.062 ± 0.000	0.390 ± 0.000	0.030 ± 0.000

Our final model combined text-based pseudo-labels generated by a two-step BERT training (first on volunteer ratings, then difficulty labels) with tabular data. Although this hybrid approach should theoretically enhance performance, results from our experiments indicated that while easy tasks showed a relatively stable AUC around 0.703, hard task performance (0.356 ROC-AUC) did not meet initial expectations, signaling the need for deeper analysis and further improvement.

4.3.4 Hyperparameter Tuning and Model Optimization

Recognizing the initial underperformance, I undertook extensive hyperparameter tuning. By systematically adjusting parameters like tree depth, learning rate, and feature selection methods for LightGBM and Random Forest, I improved results. This tuning significantly boosted our ROC-AUC scores and stability across different seeds, narrowing the performance gap between our experiments and previous research. Although achieving a perfect alignment with published

figures was challenging due to data variability, significant enhancements in ROC-AUC were realized post-tuning, reflecting rigorous model optimization.

4.4 AWS Integration (Future Work)

The upcoming stage of my research involves deploying our best-performing model on AWS, ensuring scalability, reliability, and security for practical operational use.

Outcome

Our developed hybrid model achieved promising predictive performance, poised to significantly improve volunteer experience on the Food Rescue Hero platform. The comprehensive hyperparameter tuning substantially enhanced our initial results, increasing the reliability and practical value of our models.

Issues and Reflections

6.1 Personal Learning and Development

This research offered significant exposure to advanced machine learning techniques, database management, debugging, and practical research skills. I learned valuable lessons in proactively aligning my workflow with academic expectations, balancing remote research with coursework, and optimizing productivity through structured work environments.

6.2 Challenges Encountered

Initially limited exposure to databases and machine learning posed significant hurdles. Complex debugging, time-consuming dataset preparation, managing multiple model trainings, and accurate metric extractions challenged me substantially. Moreover, balancing intensive coursework with research tasks and managing autonomy presented additional complexity. These challenges taught me crucial time management skills and fostered significant personal and professional growth.

Overall, despite obstacles, this capstone research substantially enriched my academic and practical understanding of real-world machine learning applications, offering crucial insights into research methodologies and technical collaboration in data-driven environments.

Bibliography

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